

1

1

$$z = \sum w_i x_i + b$$

$$\rightarrow z = x_1 + x_2 - 0.6$$

$$z \geq 0 \leftarrow y = 1$$

$$z < 0 \leftarrow y = 0$$

$$y(0,0) = 0$$

$$y(0,1) = 1$$

$$y(1,0) = 1$$

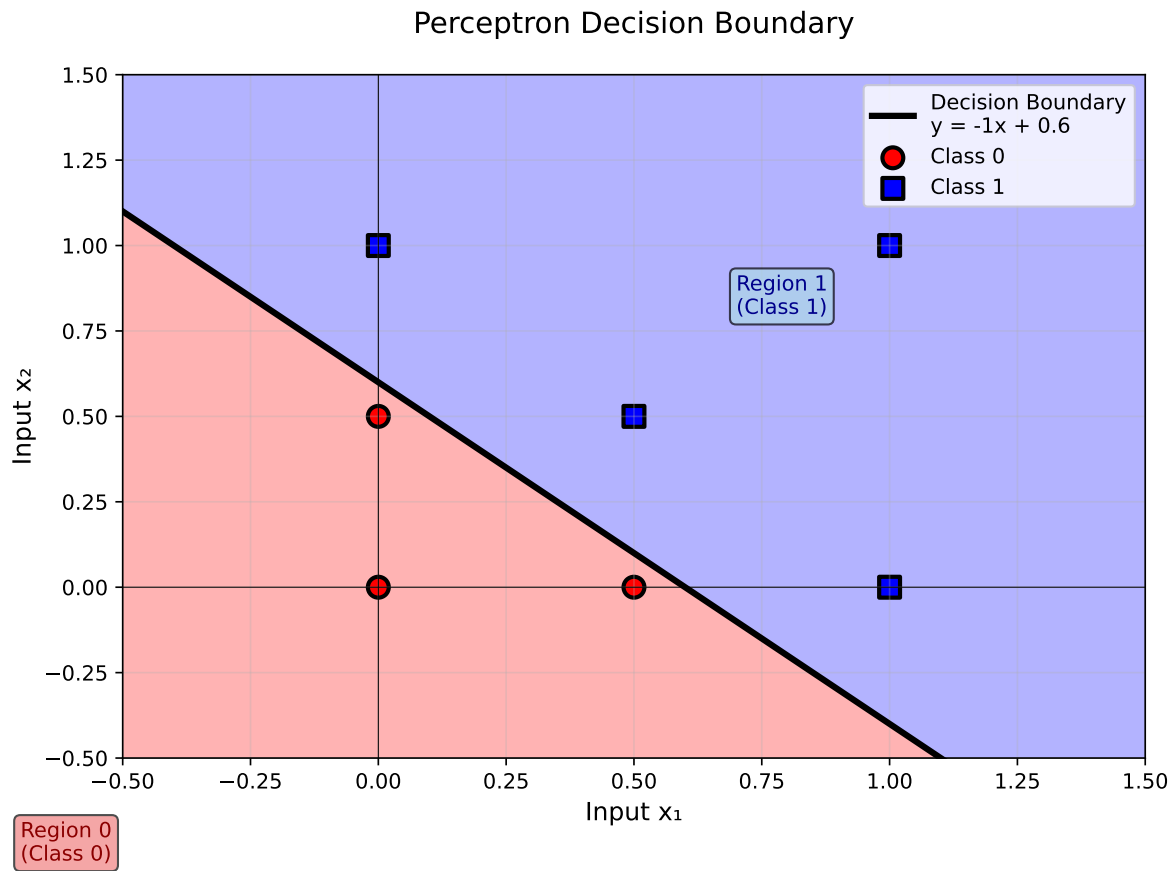
$$y(1,1) = 1$$

$$y(0,0.5) = 0$$

$$y(0.5,0) = 0$$

$$y(0.5,0.5) = 1$$

2



3

This would represent the logical OR function because the perceptron is 0 iff $x_1 = 0 \wedge x_2 = 0$

2

1 NAND

Assuming only x_1 and x_2 we get the following:

$$w_1 = -1, w_2 = -1, b = 1.5$$

We can get these weights and biases as long as they follow these inequalities:

1. $w_1 + w_2 + b < 0$
2. $w_1 + b > 0$
3. $w_2 + b > 0$
4. $b > 0$

For an n input NAND operator:

$$\forall i (w_i = -1), b = n - 0.5$$

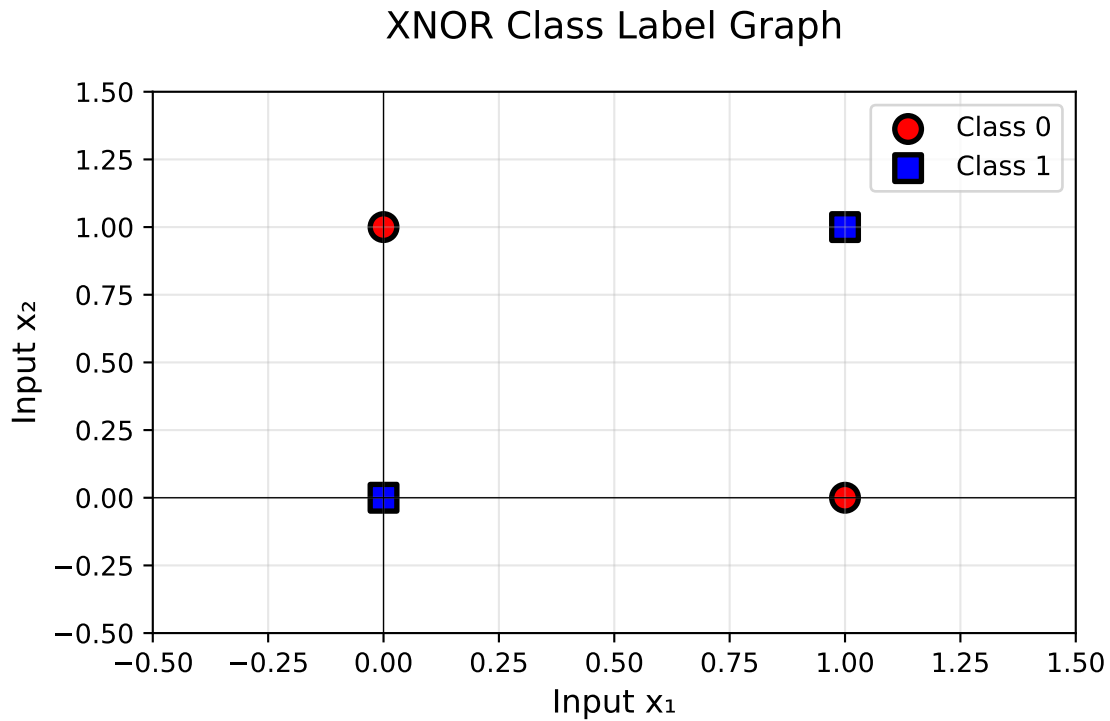
2 NOT

Assuming only x_1 we get the following: $w_1 = -1, b = 0.5$

1. $w_1 + b < 0$
2. $b > 0$

3

1



2

No single straight line can separate the two classes because the classes are in opposite corners as shown above.

3

No it is not possible to construct a perceptron for the two input XOR. We know that the weights and biases must follow the below inequalities:

1. $w_1 + w_2 + b < 0$
2. $w_1 + b > 0$
3. $w_2 + b > 0$

4. $b < 0$

From 4, we know b must be negative. Using this information we know that w_1, w_2 must both be positive to make 2,3 true.

Now looking at number 1, $w_1 + w_2 + b < 0$, we can substitute inequalities 2,3 in to get:

$$w_1 + w_2 + b > -b - b + b$$

$$\rightarrow -b - b + b = -b$$

From inequality 4, we know that $-b > 0$ so we know that $w_1 + w_2 + b > 0$ which creates a contradiction because inequality 1 says that $w_1 + w_2 + b < 0$.

Therefore it is not possible to create a perceptron for the XOR logical operation.

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